



BuchhaltungsButler

Work case

**Welcome to your work case for our AI
Software Developer position.**

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[Good luck!](#)



Work case

Greenfield architecture for agentic accounting product

BuchhaltungsButler is on a growth trajectory and is increasingly focusing on e-commerce customers with high data volumes. This means we have an increased need for automation and self-service capabilities to be able to create a high quality experience for our product.

The product today is a MVC application written in PHP Laminas with some [API capabilities](#) available. One of the bets is to mature in AI and Automation, to provide accurate booking suggestions for accounting, avoiding customers to do manual corrections during their accounting tasks.

The challenge:

Booking suggestion engine for accounting is today built on deterministic logic that misses edge cases, and difficult to maintain.

We have a rule-based engine for posting suggestions that can't handle all use cases. The current SaaS / frontend experience is also obsolete which require a rework into an AI-assisted experience. Today we are building an AI Assistant for simpler use cases that can later have tool calls.

Provide a greenfield architecture of an agentic accounting application, and also showcase how this should be served to the customer.

The current setup.

Component	Details
Compute	Classic EC2 instances. 1x App-Server (Ubuntu/Nginx) for application delivery, 2x Worker-Servers (Ubuntu) for processing background tasks. 5 EC2 instances with running docker containers for our microservices.
Database	Amazon Aurora Cluster (MySQL) with one writer and one reader instance.
Storage	S3 for document storage.
Network	DNS via Route 53, traffic distribution via ELB.
Security	IAM, Security Groups, Firewall
Monitoring / Logging	Cloud Watch, AWS Inspector, Cloud Guardian, ELK Stack (as Docker container on an EC2 instance, Tideways)
Deployment & Ops	Application deployment is automated (Jenkins). Server configuration (OS level, PHP settings) via Ansible. AWS resources (network, instances, security groups) are provisioned and managed manually.

This challenge presents a critical juncture for our AI and Automation enablement.

As a AI Software Developer, you will take technical responsibility and full ownership for our AI & Automation initiatives, together with our tech lead. Please develop a strategic concept that professionalizes our infrastructure and addresses the following three topics:

- **Agentic Architecture**

Analyze the current setup and design a target architecture for the agentic product, and how it communicates with the current core application. Explain your approach and your technology choice. When would a RAG implementation be relevant, and when would a deterministic suggestion be fine?

- **Human-in-the-loop process**

Showcase how and when human-in-the-loop step is required for the agentic workflow. How would this be implemented? What's patterns are important to implement to ensure the system is trustworthy for a human in an accounting software?

- **Trade-offs in design**

It is easy to over engineer AI solutions. In your solution it should be clear where and when to make tradeoffs in order to get a customer facing application in short time, aligning on customer expectations, and create a good enough and evolvable system.



Case instructions

These case instructions will serve as your guideline.

- In the interview, we will ask you to present your results. Please limit your presentation to max. 30 minutes.
- Feel free to choose however you want to present this, e.g. make a PowerPoint/ Google slides presentation, but limit it to 10 slides, prototype implementation, or diagrams.
- We are curious about the choices you have made when solving the challenges and will ask you questions along the way to facilitate a natural discussion around the topics.
- You have limited time and information, so we understand and expect you to take the necessary steps to solve the task accordingly. We simulate real-world working scenario. If requirements are unclear, we expect you to proactively ask questions (over email) instead of making assumptions that later lead to incorrect results.



Next steps

Case presentation:

Jan xx, xx – xx:xx am

You'll present your results in the tech team.

Work case presentation | 30 min

You'll present your case and how you approached it. We will ask you questions along the way to facilitate a natural discussion around the topics.

Question round | 15 min

After your presentation, we will have time for further questions.



**Best of luck
with the
case**